```python
from vehicles.cars import available_models as CarModels
from vehicles.bikes import available_models
print("Cars:",CarModels())
```

```
Cars: ['Ferrari', 'Mustang', 'BMW']
```

```python
import os

import os

# Get current directory
current_dir = os.getcwd()

# List all files in current directory
files = os.listdir(current_dir)

print("All files in current directory:\n")
for file in files:
    if os.path.isfile(file):
        print(file)

# Filter .py files
py_files = [file for file in files if file.endswith('.py')]

# Display count of .py files
print("\nTotal number of .py files:", len(py_files))

# Display content of one .py file
if py_files:
    print("\nDisplaying content of:", py_files[0])

    with open(py_files[0], 'r') as f:
        content = f.read()
        print("\nFile Content:\n")
        print(content)
else:
    print("\nNo .py files found in the directory.")
```

```
All files in current directory:

friends.txt
PracticalSlipSolutions.ipynb

Total number of .py files: 0

No .py files found in the directory.
```

```python
str = input("Enter the string : ")
space = c = s = v = 0
for ch in str:
    if ch == " ":
```

```
        space += 1
    elif ch.isalpha():
        if ch.lower() in 'aeiou':
            v += 1
        else:
            c += 1
    else:
        s += 1
print("Vowels :",v)
print("Consonents :",c)
print("Special Symbol :",s)
print("Spaces :",space)
```

```
import pandas as pd
import numpy as np
data = {
    "player" : ['Hardik Pandya', 'K L Rahul', 'Andre Russel', 'J
Bumrah', 'V Kohli', 'R Sharma'],
    "Team" : ['MI', 'KIP', 'KKR', 'MI', 'RCB', 'R Sharma'],
    "Category" : ['All Rounder', 'Batsman', 'Batman', 'Bowler',
'Batsman', 'Batsman'],
    "bidPrice" : [13, 12, 7, 10, 17, 15],
    "runs" : [1000, 2400, 900, 200, 3600, 3700]
}
df = pd.DataFrame(data)
print(df)
```

```
          player      Team     Category  bidPrice  runs
0  Hardik Pandya        MI  All Rounder        13  1000
1      K L Rahul       KIP      Batsman        12  2400
2   Andre Russel       KKR       Batman         7   900
3       J Bumrah        MI       Bowler        10   200
4        V Kohli       RCB      Batsman        17  3600
5       R Sharma  R Sharma      Batsman        15  3700
```

```
df.head(2)

df.tail(3)

df.isnull()

m = df['bidPrice'].mean()
print(m)
fillna(m,inplace = True)

df['bidPrice'].fillna(df['bidPrice'].mean(), inplace=True)

# User-defined Exception
class InvalidMarksError(Exception):
    pass
```

```python
try:
    # Input marks
    marks = float(input("Enter marks (0 to 100): "))

    # Check for invalid input (out of range)
    if marks < 0 or marks > 100:
        raise InvalidMarksError("Marks should be between 0 and 100.")

    # Display Grade
    if marks >= 90:
        grade = 'A+'
    elif marks >= 75:
        grade = 'A'
    elif marks >= 60:
        grade = 'B'
    elif marks >= 50:
        grade = 'C'
    elif marks >= 40:
        grade = 'D'
    else:
        grade = 'Fail'

    print("Valid Marks:", marks)
    print("Grade:", grade)

# Handle user-defined exception
except InvalidMarksError as e:
    print("Error:", e)

# Handle invalid input (non-numeric)
except ValueError:
    print("Error: Please enter a valid numeric value.")

import random

lst = [1, 2, 3, 4, 5, 6]
random.shuffle(lst)
print(lst)

fp = open('friends.txt', "r")
content = fp.read()
print("File data :",content)
fp.close()

f1 = open('friends.txt','a')
f1.write('Pratik Gadekar')
f1.close()
f1 = open('friends.txt','r')
print("Content :",f1.read())
f1.close()
```

```
Content : Akshay Shelke
Tejas Pawde
Deep Shelar
Kim
Pratham TambePratik GadekarPratik Gadekar
```

```
f1 = open('friends.txt','a')
f1.write('\nTejas Tambe')
f1.close()

f = open('friends.txt','r')
print("Content:\n",f.read())
f.close()
```

```
Content:
 Akshay Shelke
Tejas Pawde
Deep Shelar
Kim
Pratham TambePratik GadekarPratik Gadekar
Tejas Tambe
```

```
text = input("Enter a string: ")
words = text.split()
print("Words without vowels:")
for word in words:
    if not any(char.lower() in 'aeiou' for char in word):
        print(word)
```

```
Enter a string:  satyam is my name
```

```
Words without vowels:
my
```

```
text = input("Enter text")
words = text.split()
for word in words:
    if not any(char.lower() in 'aeiou' for char in word):
        print(word)
```

```
Enter text lfuwriubr iluyf jhjfhjfh aigyf
```

```
jhjfhjfh
```

```
p = pd.read_csv("emp.csv")
print(p)
```

```
    name  designation  salary
0   Tejas          HR  100000
1     Kim    Voice HR   90000
2  Prasad       Clerk   50000
```

```python
import pandas as pd

data = {'mobiles' : ['samsung S23', 'iPhone 16', 'Google Pixel'],
        'prices' : [70000, 80000, 75000]
}
df = pd.DataFrame(data)
print("Mobile Dataframe")
print(df)
```

```
Mobile Dataframe
        mobiles  prices
0   samsung S23   70000
1     iPhone 16   80000
2  Google Pixel   75000
```

```python
from vehicles1 import car, bike
print("Cars :",car.available_models())
print("Bikes :",bike.available_models())
```

```
Cars : ['Ferrari', 'BMW', 'Tata Nexon', 'Scorpio']
Bikes : ['MT15', 'Ninja', 'Splender', 'Pulser']
```

```python
str = input("Enter the string")
v = c = s = 0;
for ch in str:
    if ch.isalpha():
        if ch.lower() in 'aeiou':
            v += 1
        else:
            c += 1
    else :
        s += 1
print("Vowels :",v)
print("Consonents :",c)
print("Special Symbol :", s)
```

```
Enter the string satyam thorat
```

```
Vowels : 4
Consonents : 8
Special Symbol : 1
```

```python
import pandas as pd
import numpy as np
IPL = {
    'players' : ['hardik', 'KL Rahul', 'Russel', 'Bumrah', 'VK',
'Rohit Sharma'],
    'team' : ['MI', 'KE', 'KKR', 'MI', 'RCB', 'MI'],
    'category' : ['bat', 'bat', 'bat', 'bow', 'bat', 'bat'],
    'bidPrice' : [13, 12, np.nan , 10, 17, np.nan],
    'runs' : [1000, 2400, 900, 200, 3600, 3700]
```

```
}
df = pd.DataFrame(IPL)
print(df)
```

```
        players team category  bidPrice  runs
0        hardik   MI      bat      13.0  1000
1      KL Rahul   KE      bat      12.0  2400
2        Russel  KKR      bat       NaN   900
3        Bumrah   MI      bow      10.0   200
4            VK  RCB      bat      17.0  3600
5  Rohit Sharma   MI      bat       NaN  3700
```

```
df.head(3)
```

```
    players team category  bidPrice  runs
0    hardik   MI      bat      13.0  1000
1  KL Rahul   KE      bat      12.0  2400
2    Russel  KKR      bat       NaN   900
```

```
exp = df['bidPrice'].idxmax()
exp
```

```
4
```

```
exp = df.loc[df['bidPrice'].idxmax()]
exp
```

```
players       VK
team         RCB
category     bat
bidPrice    17.0
runs        3600
Name: 4, dtype: object
```

```
count = df['team'].value_counts()
print(count)
```

```
team
MI     3
KE     1
KKR    1
RCB    1
Name: count, dtype: int64
```

```
df['bidPrice'] = df['bidPrice'].fillna(df['bidPrice'].mean())
df
```

```
        players team category  bidPrice  runs
0        hardik   MI      bat      13.0  1000
1      KL Rahul   KE      bat      12.0  2400
2        Russel  KKR      bat      13.0   900
3        Bumrah   MI      bow      10.0   200
4            VK  RCB      bat      17.0  3600
5  Rohit Sharma   MI      bat      13.0  3700
```

```
df.head(2)
```

```
    players team category  bidPrice  runs
0    hardik   MI      bat      13.0  1000
1  KL Rahul   KE      bat      12.0  2400
```

```
df.tail(3)
```

```
        players team category  bidPrice  runs
3        Bumrah   MI      bow      10.0   200
4            VK  RCB      bat      17.0  3600
5  Rohit Sharma   MI      bat       NaN  3700
```

```
df.isnull()
```

```
   players   team  category  bidPrice   runs
0    False  False     False     False  False
1    False  False     False     False  False
2    False  False     False      True  False
3    False  False     False     False  False
4    False  False     False     False  False
5    False  False     False      True  False
```

```
df['bidPrice'] = df['bidPrice'].fillna(df['bidPrice'].mean())
print(df)
```

```
        players team category  bidPrice  runs
0        hardik   MI      bat      13.0  1000
1      KL Rahul   KE      bat      12.0  2400
2        Russel  KKR      bat      13.0   900
3        Bumrah   MI      bow      10.0   200
4            VK  RCB      bat      17.0  3600
5  Rohit Sharma   MI      bat      13.0  3700
```

```
df.max()
```

```
players     hardik
team           RCB
```

```
category       bow
bidPrice      17.0
runs          3700
dtype: object
```

```
df.max()
```

```
players     hardik
team           RCB
category       bow
bidPrice      17.0
runs          3700
dtype: object
```

```
most_expensive = df.loc[df['bidPrice'].idxmax()]
print(most_expensive)
```

```
players       VK
team         RCB
category     bat
bidPrice    17.0
runs        3600
Name: 4, dtype: object
```

```
most_expensive = df.loc[df['bidPrice'].idxmin()]
print(most_expensive)
```

```
players     Bumrah
team            MI
category       bow
bidPrice      10.0
runs           200
Name: 3, dtype: object
```

```
df['bidPrice'] = df['bidPrice'].fillna(df['bidPrice'].mean())
print(df)
```

```
        players team category  bidPrice  runs
0        hardik   MI      bat      13.0  1000
1      KL Rahul   KE      bat      12.0  2400
2        Russel  KKR      bat      13.0   900
3        Bumrah   MI      bow      10.0   200
4            VK  RCB      bat      17.0  3600
5  Rohit Sharma   MI      bat      13.0  3700
```

```
print(df['team'].value_counts())
```

```
team
MI     3
KE     1
KKR    1
```

```
RCB    1
Name: count, dtype: int64
```

```python
count = df['team'].value_counts()
print(count)
```

```
team
MI     3
KE     1
KKR    1
RCB    1
Name: count, dtype: int64
```

```python
count = df['team'].value_counts()
print(count)
```

```
team
MI     3
KE     1
KKR    1
RCB    1
Name: count, dtype: int64
```

```python
teamm = df['team'].value_counts()
print(teamm)
```

```
team
MI     3
KE     1
KKR    1
RCB    1
Name: count, dtype: int64
```

```python
import re
```

Q1. Write a Python program that accepts value of m and n. Divides m by n and handles a ZeroDivisionError if the user tries to divide by zero.

```python
try:
    m = int(input("Enter numorator "))
    n = int (input("Denominator "))
    d = m/n
    print(d)
except ZeroDivisionError:
    print("Canot Divide by Zero")
```

```
Enter numorator  2
Denominator  0

Canot Divide by Zero
```

Write a Python program to demonstrate a user-defined exception that checks whether the entered marks are between 0 and 100. Display an appropriate message for invalid input and out-of-range values. Also display Grade of marks.

```
class InvalidMarksError(Exception):
    pass
try:
    print
    marks = int(input("Enter marks"))
    if marks < 0 or marks > 100:
        raise InvalidMarksError("marks should between 0 and 100")
    if marks >= 90 and marks < 100:
        print("A+")
    elif marks >=80 and marks < 90:
        print("A grade")
    elif marks >= 70 and marks <80:
        print("B+ grade")
    elif marks >= 60 and marks <70:
        print("B grade")
    elif marks >=45 and marks <60:
        print("C grade")
    elif marks >=0 and marks <45:
        print("Fail")
except ValueError:
    print("Marks must be integer")
except InvalidMarksError as e:
    print("Error:",e)
```

```
Enter marks d
```

```
Marks must be integer
```

Write a python program to create a module containing factorial function, power dictionary and list of vowels. Save this as variables.py. Import variables module to print the factorial of n, print the power of n, and second alphabet from the vowels list. (Accept n from user)

```
filename = input("Enter file name: ")

try:
    f = open(filename, 'r')
    content = f.read()

    print("\nFile Content:\n", content)

    print("Characters:", len(content))
    print("Words:", len(content.split()))
    print("Lines:", len(content.splitlines()))

    f.close()
```

```
except FileNotFoundError:
    print("File does not exist!")
```

```
Enter file name:  friends.txt


File Content:
 Akshay Shelke
Tejas Pawde
Deep Shelar
Kim
Pratham TambePratik Gadekar
Tejas Tambe
Characters: 81
Words: 12
Lines: 6
```

```
import variables as v
n = int(input("enter"))
print('factorial',v.factoria(n))
print('Power :',v.power[4])
print('Vowels :',v.vowels[2])
```

```
enter 4

factorial 24
Power : 16
Vowels : i
```

```
import random

import random
list = [1, 2, 3, 4, 5]
random.shuffle(list)
print("Shuffeld",list)
```

```
Shuffeld [4, 1, 3, 2, 5]
```

```
import pandas as pd

data = {
    "Company": ["Apsara", "Natraj", "Cello", "Parkar", "Apsara"],
    "Count": [15, 20, 25, 35, 20],
    "Price": [250, 200, 600, 900, 300]
}

index = ["Pencil", "Pencil", "Pen", "Pen", "Eraser"]

df = pd.DataFrame(data, index=index)
```

```python
# a
print(df.loc["Pencil"])

# b
df.loc["Eraser", "Count"] = 25

# c
print(df[["Company", "Price"]])

# d
print(df.loc[["Pencil", "Pen"]])

# e
df.rename(columns={"Count": "Quantity"}, inplace=True)
print(df)
```

```
       Company  Count  Price
Pencil  Apsara     15    250
Pencil  Natraj     20    200
       Company  Price
Pencil  Apsara    250
Pencil  Natraj    200
Pen      Cello    600
Pen     Parkar    900
Eraser  Apsara    300
       Company  Count  Price
Pencil  Apsara     15    250
Pencil  Natraj     20    200
Pen      Cello     25    600
Pen     Parkar     35    900
       Company  Quantity  Price
Pencil  Apsara        15    250
Pencil  Natraj        20    200
Pen      Cello        25    600
Pen     Parkar        35    900
Eraser  Apsara        25    300
```

```python
with open('friend.txt','w') as f:
    f.write("Abhay\n")
    f.write("Kiran\n")
    f.write("Prashant\n")
    f.write("Ashwin\n")
    f.write("Dom\n")
    f.close()

with open('friend.txt','r') as f:
    print(f.read())
    f.close()
```

```
Abhay
Kiran
```

```
Prashant
Ashwin
Dom
```

```python
with open('friend.txt', 'a') as f:
    f.write("Rajaram")
    f.close()
```

```python
with open('friend.txt','r') as f:
    print(f.read())
```

```python
from banking import loan, account
account.open_account("Satyam")
loan.apply_loan(10000)
```

```
Account created  Satyam
Applied for loan  10000
```

```python
str = input("enter string")
words = str.split()
result = []
for w in words:
    has_vowel = False
    for ch in w:
        if ch.lower() in 'aeiou':
            has_vowel = True
            break
    if not has_vowel:
        result.append(w)
print("Words without vowel :",result)
# print(words)
```

```
enter string asd ddl bdl mst satyam
```

```
Words without vowel : ['ddl', 'bdl', 'mst']
```

```python
s = input("Enter string")
words = s.split()
result = []
for w in words:
    has_vowel = False
    for ch in w:
        if ch.lower() in 'aeiou':
            has_vowel = True
            break
    if not has_vowel:
        result.append(w)
print("Words without vowels :",result)
# print(words)
```

```
Enter string satyam dld hgrt nhca loiu bh rtd

Words without vowels : ['dld', 'hgrt', 'bh', 'rtd']
```

Write a Python program that prompts the user to input two numbers and raises a TypeError exception if the inputs are not numerical.

```
try:
    a = int(input("Enter num1"))
    b = int(input("Enter num2"))
    print("sum :",a+b)
except ValueError:
    print("Numbers must be integer")
```

```
Enter num1 3
Enter num2 f

Numbers must be integer
```

```
try:
    a = input("Enter num1: ")
    b = input("Enter num2: ")
    if not (a.isdigit() and b.isdigit()):
        raise TypeError("Numbers Must be integer")

    a = int(a)
    b = int(b)
    print("Sum:", a + b)
except TypeError as e:
    print("Error :",e)
```

```
Enter num1:  f
Enter num2:  3

Error : Numbers Must be integer
```

```
try:
    a = input("Enter first number: ")
    b = input("Enter second number: ")

    if not (a.isdigit() and b.isdigit()):
        raise TypeError("Inputs must be numbers")

    a = int(a)
    b = int(b)

    print("Sum:", a + b)
```

```
except TypeError as e:
    print(e)
```

```
Enter first number:  3
Enter second number:  4

Sum: 7
```

Create a following DataFrame named as “data”. Write the python code for the following commands.

```
import pandas as pd
data = pd.DataFrame({
    "Product" : ['pencil', 'pencil', 'pen', 'pen', 'eraser'],
    'Brand': ['Apsara', 'Natraj', 'Cello', 'Parkar', 'Apsara'],
    'Count': [15, 20, 25, 35, 20],
    'Price': [250, 200, 600, 900, 300]
})
data
```

```
  Product   Brand  Count  Price
0  pencil  Apsara     15    250
1  pencil  Natraj     20    200
2     pen   Cello     25    600
3     pen  Parkar     35    900
4  eraser  Apsara     20    300
```

```
print(data[data['Product'] == 'pencil'])
```

```
  Product   Brand  Count  Price
0  pencil  Apsara     15    250
1  pencil  Natraj     20    200
```